Maths Exploring at Terminal-03

Progressive exercises

E1 · Direct application. We define on $\mathbb{N}$ the follow-up (u_n) and (v_n) by $u_n = \frac{4n-1}{n+2}$ and $v_n = n^2 - 7n$. Determine their limit by justifying the transformations used.

E2 · Application and justification. We define $u_n = \sqrt{n+1} - \sqrt{n}$ for every $n \in \mathbb{N}$. Using the the combined quantity, determine the limit of (u_n).

E3 · Method to be organised. We define $u_n = \frac{\cos n}{n}$ for the integer $n \geq 1$. Framework u_n, and then deduct from it limit of (u_n).

E4 · Choice of strategy. Next (u_n) is increasing and verifies $u_n < 5$ for all $n \in \mathbb{N}$. Justify that Is it necessarily equal to 5? Justify.

E5 · Synthesis or demonstration. We define $u_n = 3 - 2(-0{,}6)^n$ for every $n \in \mathbb{N}$. Determine the limit of (u_n). Is this sequence monotonous?